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A DFT and microkinetic study of HCHO catalytic oxidation mechanism over Pd/Co_3O_4 catalysts: The effect of metal-oxide interface

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ABSTRACT

The oxidation of HCHO at the interface between Pd and Co_3O_4 support was studied by using density of functional theory (DFT) calculation. A Pd/Co_3O_4 model was established to investigate the complex interaction of reactants, products, and catalysts under reaction conditions. The interface interaction led to the electron migration from Pd to Co_3O_4, which enhanced the adsorption of HCHO and O_2. Pd atoms played the role as the active sites in the whole process of HCHO oxidation. Interface interaction changed the adsorption mode of CO oxidation, which contributed to CO activation. The calculated results showed that the energy barrier of CO oxidation at Pd/Co_3O_4 interface was lower than that on Pd (111) surface. Barrier decomposition analysis showed CO activation determined the energy barrier difference between the interface and Pd (111) surface. Microkinetics studies indicated that CO oxidation on Pd (111) surface was the rate limiting step, and the catalyst surface was covered with CO molecules. At Pd/Co_3O_4 interface, CO was completely converted to CO_2 and the adsorption of HCHO and O_2 was the rate limiting step at low temperature. At Pd/Co_3O_4 interface, the production rate of CO_2 and H_2O was 15 times higher than that on Pd (111) surface, which indicated that Pd/Co_3O_4 interface could improve catalytic activity of HCHO, especially at room temperature.

1. Introduction

Formaldehyde (HCHO) has been regarded as one of the major toxic indoor pollutants, which may cause severe health problems such as headache, nasal tumors, eye and skin irritation, respiratory tract [1]. Therefore, great efforts have been made to remove HCHO from indoor air, e.g. physical and chemical adsorption [2–4], plasma technology [5], photocatalytic degradation [6–8], and catalytic oxidation [9,10]. Complete catalytic oxidation of HCHO to harmless CO_2 and H_2O is considered as a promising technique for HCHO abatement due to its low cost, high removal efficiency and eco-friendly reaction conditions [11,12].

Generally, there are two major types of efficient catalysts for HCHO oxidation: supported noble metals and transition metal oxides. Cheap transition metal oxides, such as Co_3O_4, MnO_x, and CeO_2, may be a good substitute for noble metal. However, the low efficiency of these catalysts can not satisfy the needs of complete HCHO elimination at low temperature. Noble metal catalysts supported on transition metal oxides exhibit excellent catalytic activity for HCHO oxidation at room temperature, which have drawn lots of attention in recent years [12–14]. Huang et al. [15] confirmed that the reduced Pt/TiO_2 catalysts with metallic Pt showed much higher HCHO conversion than the unreduced ones with cationic Pt in all cases, suggesting that the oxidation state of Pt was crucial to the catalytic activity. Similar results were obtained on Pd/TiO_2 catalyst [9]. However, some researches have showed that the activity of Au^{3+} was higher than that of Au° [16,17]. Therefore, the debate on oxidation states of noble metals still exists. On the other hand, the role of supports in the catalytic reaction over metal-oxide is not clear. Peng et al. [18] investigated the catalytic activities of Pt on a series of supports for HCHO oxidation and found that the catalytic activity was closely related to the dispersion of platinum on supports rather than the specific surface areas of supports. The high dispersion of platinum can provide more catalytic sites. However, Grisel and Nieuwenhuys [19], who studied the effect of the support over $Au/MO_x/Al_2O_3$ (M = Cr, Mn, Fe, Co, Ni, Cu, Zn), indicated that the catalytic activity highly depends on both the particle size and the type of MO_x. It has also been claimed that the support is directly involved in the reaction when reducible oxides are used as the support. More importantly, due to metal–oxide interface effect [20–24], the mobility of the lattice oxygen, the reducibility of catalyst, electron transfer, oxygen adsorption and activation are all so complex. Thus, understanding the nature of metal–oxide interaction is quite challenging.

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In this study, a typical Pd/Co_3O_4 model was constructed to investigate the interface between metal and oxide. Co_3O_4 was used as a support for its relative low toxicity and strong stability and oxidation ability [25–27], and noble metal supported on Co_3O_4 for HCHO oxidation has been reported by many experiments [27–31]. However, the atomic details of interface effect are sitll elusive, which provides an impetus for the study of interaction between Pd and Co_3O_4 support by combining DFT calculations with microkinetics simulations.

2. Methods and models

2.1. DFT calculations

All spin-polarized calculations were performed with Vienna Ab Initio Simulation Package (VASP) [32] using the GGA-PBE electron exchange-correlation functional [33]. The projector augmented wave (PAW) method was used to describe these orbitals in the core regions [34]. 1 × 1 × 1 *k*-point mesh and kinetic energy cutoff of 400 eV were used. The structures were relaxed until the forces acting over all atoms were less than 0.05 eV/Å, and electronic energies were converged in the range of 10^{-4} eV. The climbing-image nudged elastic band module was employed to locate the transition states [35]. Transition states were verified to be a first-order saddle point by numerical frequency analysis. The adsorption energy of surface species is calculated by

$$E_{ads}(x) = E(sur) + E(x) - E(x + sur)$$

Where x, $E_{ads}(x)$, $E(sur)$, $E(x)$ and $E(x + sur)$ represent the surface species, the adsorption energy of x, the energy of clean surface and the energy of x and the energy of the surface covered by x, respectively.

2.2. Microkinetics simulations

MKMCXX software is used for microkinetics simulation [36,37]. For surface reactions, the rate constant of the forward (reverse) reactions can be derived from the Eyring equation:

$$k = Ae^{-\frac{E_a}{k_b T}}$$

where A, k, Ea, k_b, and T are the prefactor, the reaction rate constant, the activation barrier, Boltzmann constant temperature in Kelvin, respectively.

For non-activated molecular, the gas species adsorption and desorption rates are given by the Hertz-Knudsen kinetics [38]:

$$k_{ads} = S\frac{PA}{\sqrt{2\pi m k_b T}}$$

$$k_{des} = A\frac{k_b T^3}{h^3}\frac{2\pi k_b}{\sigma\theta_{rot}}\exp\left(\frac{-E_{des}}{RT}\right)$$

$$r = k_{ads}\cdot p_A - k_{des}\cdot[A_{ads}]$$

Where S, A, m, σ,? $?_{rot}$, and P stand for the sticking coefficient, the area of the surface site, the mass of the molecule, the symmetry number, the characteristic temperature for rotation, and the partial pressure of adsorbate in the gas, respectively.

The differential equations for the surface reaction intermediates are given by the rate constants and elementary reaction steps. A single differential equation for M components is written in the form:

$$r_i = \sum_{j=1}^{N}\left(k_j v_i^j \prod_{k=1}^{M} c_k^{v_k^j}\right)$$

Here v_i is the stoichiometric coefficient of component i in reaction step k; kj is the reaction rate constant; c_k is the concentration of component k on the surface.

The degree of rate control (X_{RC}) is calculated to measure the variance of the overall rate on slight changes in the energy of transition state of step i:

$$X_{RC,i} = \frac{k_i}{r}\left(\frac{\partial r}{\partial k_i}\right)_{k_{j\neq i},K_i} = \left(\frac{\partial \ln r}{\partial \ln k_i}\right)_{k_{j\neq i},K_i}$$

Where k_i, K_i and r respent the rate constant, the equilibrium constant and the reaction rate, respectively.

2.3. Models

Co_3O_4 (110) surface, which is the main surface on Co_3O_4 nanorods [39,40], has two different terminations: A and B. Co_3O_4 (110)-B terminal was chosen to model the support and was simplified to Co_3O_4, because the (110)-B surface was energetically more stable than the (110)-A surface under the common working conditions [39,41]. U_{eff} = 3 was set to provide a better description of the electronic structure and surface property [42].

The Co_3O_4 (110) surface is represented with a p(2 × 2) unit cell. The slab is five layers thick (two layers at the bottom were frozen and the other atoms were completely relaxed) and is separated from periodic images by 15 Å of vacuum spacing. In order to construct Pd model, the lattice constant of bulk Pd is optimized to 3.95, which is in consistent with the previous work. Then, a three-layer thick nanowire with two or three atomic wide is cut from Pd (2 × 2) supercell. Pd/Co_3O_4 model was constructed by fully relaxing Pd nanowire on Co_3O_4 (110) surface in the direction of the shorter cell vector. For the relaxation of Pd nanowire, we first kept the optimized Pd atoms relaxing only in the vertical direction (Fig. 1a). Vertical relaxation of Pd nanowire allows the incorporation of changes in the metal/support adhesion caused by the modified interfacial dipoles due to charge transfer [43]. An approach similar to the one proposed by Schneider et al [43] was tested, where we took the structure obtained in Fig. 1a and allowed unconstrained relaxation of Pd along the interface while keeping the other Pd atoms fully fixed. The optimized geometry is shown in Fig. 1b. Based on the model in Fig. 1b, the surface layer of Pd (dark area in Fig. 1c) was relaxed due to HCHO oxidation involved Pd (111) surface. Finally, we chose the model in Fig. 1c as Pd/Co_3O_4 model, whose energy was only 0.1 eV different from the geometry in Fig. 1a. The lattice constant for bulk Pd was calculated to 3.95, which is in good agreement with previous work [43]. The optimized lattice parameters are 5.59 and 5.75 Å for Pd (111) and Co_3O_4 (110) surface, respectively. Lattice mismatch was 2.8%.

3. Result and discussion

Pd/Co_3O_4 was taken to give a detailed description of the interface effect on HCHO oxidation. For comparison, Co_3O_4 (110) and isolated Pd (111) surface were built and shown in Fig. 2. In this work, Pd at the interface and isolated Pd (111) surface are noted as Pd surface and Pd (111) surface, respectively. First, the charge density difference is carried out to explore the electron transfer and the interaction between Co_3O_4 and Pd. It can be seen from Fig. 3 that there is a strong electrons transfer at the interface, and the interaction between Co_3O_4 and Pd atoms at the bottom layer is stronger than those on the top two layers. The calculated bader charge indicates that there is an obvious electrons transfer (0.1 - 0.2 e) from Pd atoms at the bottom layer to Co_3O_4 surface. It should be noted that the electrons of Pd atom at the top two layers slightly increase (0.01 -0.07 e from bader charge), indicating that Pd is negatively charged. Then, HCHO adsorption, O_2 adsorption and dissociation and the reaction path are investigated individually. By comparison, the corresponding structures and energies are similar with that in the case of HCHO and O_2 coadsorption.

3.1. HCHO adsorption

The optimized structures of HCHO adsorption are presented in

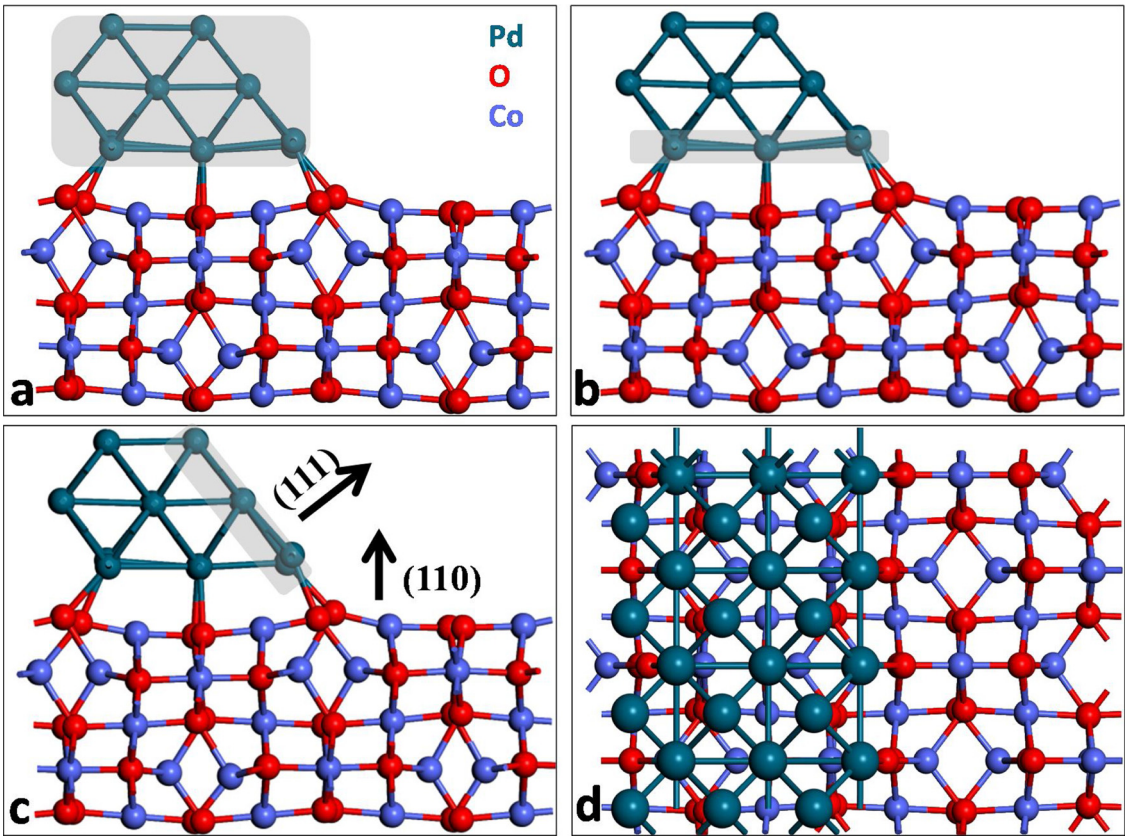


Fig. 1. Optimized geometries of Pb/Co_3O_4 model. (a) only vertical relaxations of Pd atoms are allowed; (b) metal atoms along the boundary are unconstrained, while other metal atoms are frozen in the positions obtained in (a); (c) the surface layer of Pd were relaxed and the rest were constrained; (d) the top view of geometries in (c). (The relaxed area is shown in dark).

Fig. 4. Different from the adsorption structures on the Co_3O_4 (110) and Pd (111) surface, HCHO molecules are adsorbed at the interface with C and O atoms of the carbonyl group coordinating to Pd and Co atoms, respectively. The adsorption energy of HCHO on the interface is 0.82 eV, which is lower than that on Co_3O_4 surface (1.33 eV) and higher than that on Pd surface (0.50 eV). Under low HCHO concentrations, HCHO molecules may prefer to be adsorbed on the support rather than on the interface area. However, HCHO molecules will not be further oxidized, because O_2 can not adsorb on Co_3O_4 surface. At the interface, the adsorption of HCHO leads to the conversion of C–O double bond to C–O single bond, and C–O bond extends from 1.10 Å (HCHO in the gas) to 1.29 Å. Part of electrons transfer from O and C atoms of HCHO to Co and Pd atoms.

3.2. O_2 adsorption and dissociation

Four possible adsorption configurations of O_2 at Pd/Co_3O_4 interface are shown in Fig. 5. The oxygen bridge site adsorption energy of a, b and c configuration at the interface are 0.70, 0.80, and 0.88 eV, respectively. By comparison, oxygen can be adsorbed on the bridge site of Pd (111) surface with an energy of 0.72 eV (Fig. 6a). Based on the above electronic analysis, it can be concluded that the increase of Pd valence electrons leads to the weak oxygen adsorption. The more stable adsorption mode is hollow site adsorption (Fig. 5d), and the adsorption energy is 0.96 eV. On Pd (111) surface, the adsorption energy of O_2 on hollow site is 0.80 eV (Fig. 6b). Due to the interaction between metal and oxide, the electrons on Pd atoms are transferred to Co_3O_4 support,

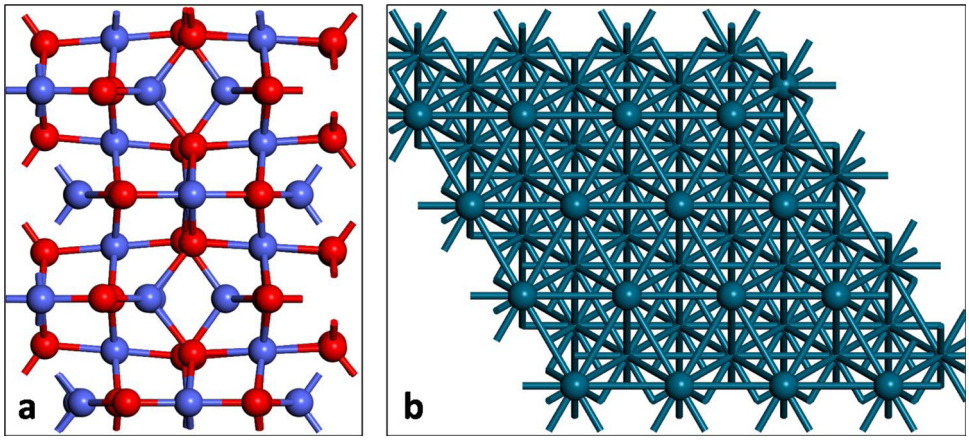


Fig. 2. Optimized geometries of (a) Co_3O_4 (110) and (b) Pd (111) surface.

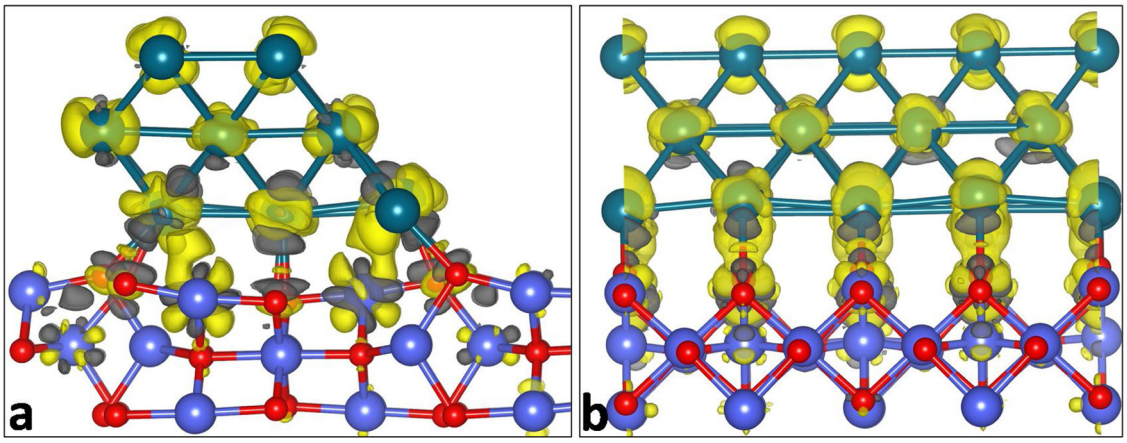


Fig. 3. (a) Side and (b) top views charge density difference for Pd/Co_3O_4. The grey and yellow volumes represent zones of electron depletion and accumulation, respectively (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

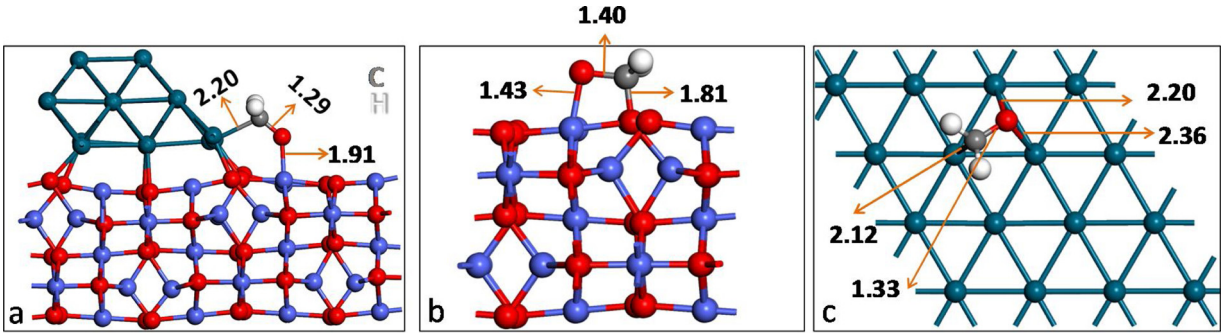


Fig. 4. HCHO adsorption structures on (a) Pd/Co_3O_4 interface, (b) Co_3O_4 (110) surface, (c) Pd (111) surface. The corresponding adsorption energies of HCHO are 0.82, 1.33 and 0.50 eV for (a), (b) and (c).

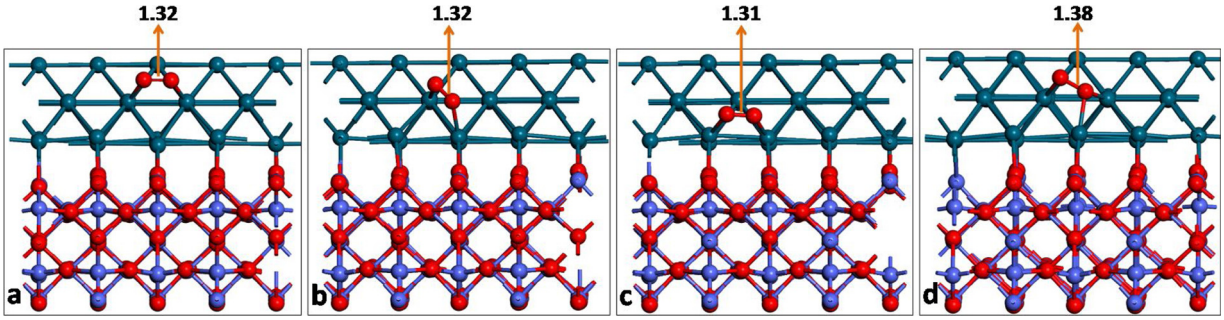


Fig. 5. Possible O_2 adsorption configuration at Pd/Co_3O_4 interface. The corresponding adsorption energies of O_2 are 0.70, 0.80, 0.88 and 0.96 eV for (a), (b), (c) and (d).

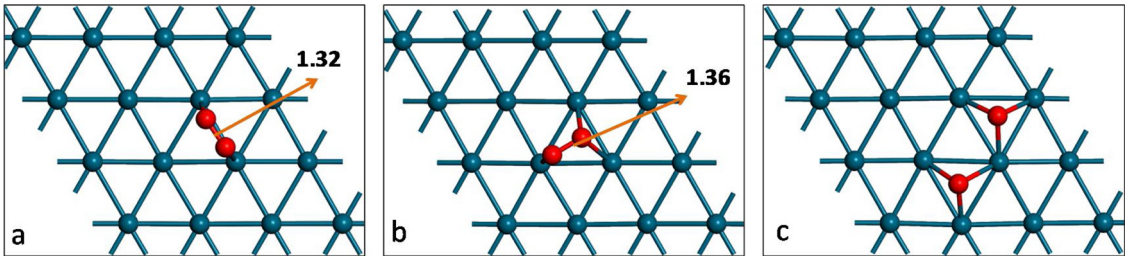


Fig. 6. O_2 adsorption and dissociation on Pd (111) surface. (a) bridge site adsorption, (b) hollow site adsorption, (c) O_2 dissociation. The corresponding adsorption energies of O_2 are 0.74 and 0.80 eV for (a) and (b). The dissociation energy barrier of O_2 in (c) is 0.74 eV.

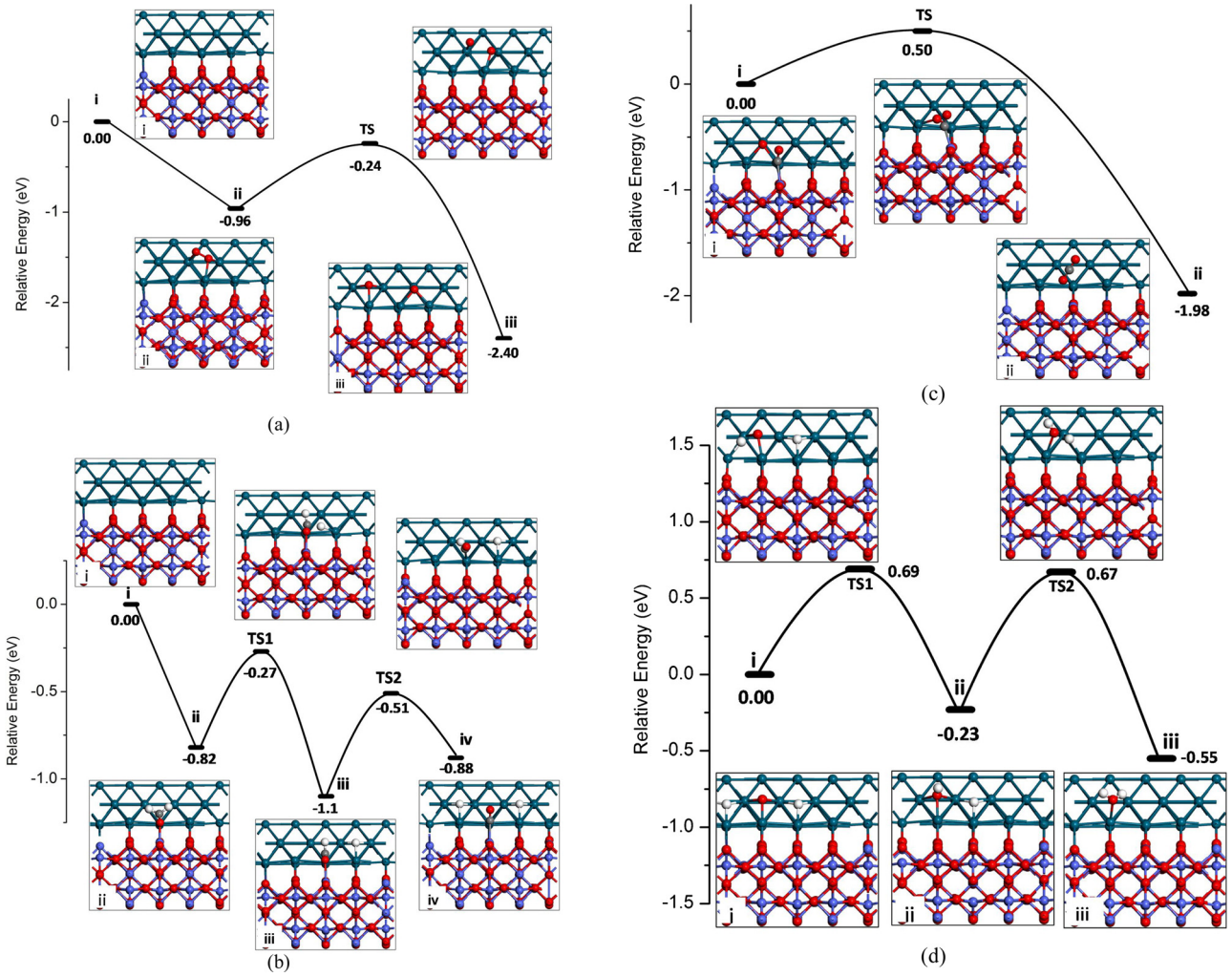


Fig. 7. The proposed reaction pathways for (a) O_2 adsorption and dissociation, (b) HCHO adsorption, HCHO → CHO + H and CHO → CO + H, (c) CO + O → CO_2, (d) H_2O formation at Pd/Co_3O_4interface.

which results in stronger oxygen adsorption. Oxygen can be further dissociated into two O atoms (Fig. 6c) with an energy barrier of 0.74 eV. The path of oxygen dissociation at the interface is shown in Fig. 7a, and the energy barrier is 0.72 eV. The results show that the dissociation of oxygen at the interface is similar to that on Pd (111) surface.

3.3. Reaction path of HCHO oxidation at Pd/Co_3O_4 interface

The proposed reaction paths of HCHO oxidation at the interface are shown in Fig. 7, including important intermediates and transition-state structures. After HCHO is adsorbed at the interface (Fig. 7b-ii), one of the C—H bonds dissociates and H atom transfers to the hollow site of Pd surface. HCHO molecule is converted into CHO group, and adsorbed at the interface (Fig. 7b-iii). The configuration of transition state is shown as TS-1 and the barrier is 0.55 eV. The step of C—H dissociation generates two redundant electrons, some of which occupy Pd atoms around CHO group, resulting in the change of magnetic moment. The other electrons are carried to Pd surface by H atom. By comparison, the cleavage barrier of C–H bond on Pd (111) and Co_3O_4 (110) surface are 0.52 and 1.37 eV, respectively. The dissociation paths are shown in Figs. 8a and 9 a. It can be found that both the interface and Pd (111) surface exhibit greater ability to activate and dissociate C–H bond than Co_3O_4 support. Then, C–H bond of CHO group can be further dissociated to form CO and H (Fig. 7b-iv). CO molecule is adsorbed at the interface and H atom is adsorbed on Pd surface. The forming CO molecule binds with Pd and Co atom, and C-Pd and C-Co bond lengths are 2.16 and 2.20 Å, respectively. The reaction energy barrier of this step is 0.59 eV. In contrast, the C–H dissociation energy barrier of CHO on Co_3O_4 (110) and Pd (111) surface are 1.07 and 0.61 eV, respectively. On Pd/Co_3O_4 interface and Pd (111) surface, the C–H dissociation path and energy of CHO group are similar. Both the interface and Pd (111) surface exhibit better catalytic performance on C–H bonds cleavage than Co_3O_4. It should be noted that CO is located on the top site of Pd atoms at the interface, while CO molecule adsorbs on the hollow site of Pd atoms on Pd (111) surface (Fig. 9b-FS). Finally, CO molecule can react with dissociated O at the interface to form CO_2 with an energy barrier of 0.50 eV (Fig. 7c). By comparison, CO molecule is adsorbed on the hollow site of Pd (111) surface and the energy barrier of forming CO_2 is as high as 1.14 eV. It can be deduced that the interaction at the interface changes the mode of CO adsorption, which contributes to the decreases of CO_2 formation energy barrier. On the surface of pure Co_3O_4 (110), O_2 can be hardly adsorbed and CO molecule can only combine with lattice oxygen atom to form CO_2.

After the cleavage of C–H bond, Pd sites around CO molecule are occupied by H atoms. In order to combine with dissociated O atom to form CO_2, the migration of H on Pd surface must be easy to achieve. The calculated result shows that H atom can be transferre on Pd surface, which needs to overcome an energy barrier of 0.06 eV. After CO_2 desorption, two H atoms and one O atom leave on the surface. H atoms can bind with O to form OH and H_2O (Fig. 7d). The corresponding energy barriers are 0.69 and 0.9 eV, respectively. The whole catalytic reaction

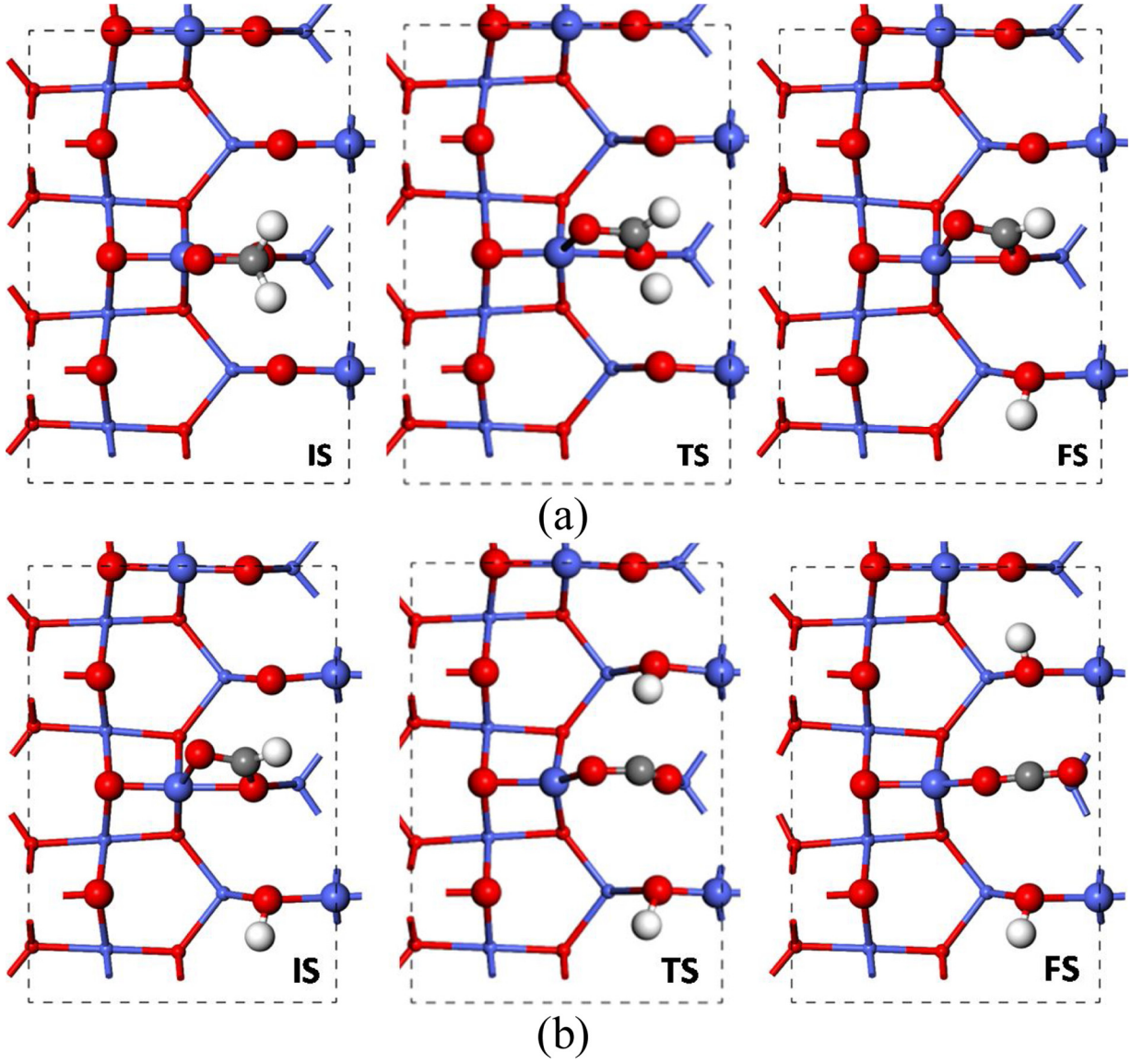


Fig. 8. C—H bond cleavage on Co_3O_4 (110) surface. (a) HCHO → CHO + H, (b) CHO → CO + H. (IS, TS, FS stand for initial state, transition state, final state).

cycle of HCHO + O_2 → H_2O + CO_2 finishes and the interface recovers.

Based on the above results, it can be concluded that the reaction path of HCHO oxidation at the Pd/Co_3O_4 interface is different from that on Co_3O_4 (110) surface. The intermediate CO binds with adsorbed oxygen species to form CO_2 rather than lattice oxygen atoms. Compared with Co_3O_4 (110) surface, the cleavage barrier of C—H bond at the interface decreases sharply. The reason is that Pd atoms play the role of active sites, and Pd metal has a better catalytic performance of HCHO oxidation than Co_3O_4 support. Compared with that on Pd (111) surface, the interaction at the interface has less impact on the steps of C–H bonds cleavage, because they mainly occur on Pd surface instead of the interface. However, the interface effect leads to the change of CO oxidation. Firstly, the metal-oxide interaction decreases the electrons of Pd at the interface, which is favor to oxygen adsorption. Secondly, the interface effect changes CO adsorption mode. At the interface, CO molecule is adsorbed on the top site of Pd surface, and binds with metal Pt and Co_3O_4 support. By contrast, CO molecule is adsorbed on hollow site of Pd (111) surface. Compared with 0.11 eV at the interface, the corresponding chemisorption energy of CO on Pt (111) surface is 2.0 eV, which is in accordance with the results in literature [44]. CO movement from the hollow site to the off-top site (from the initial state to the TS) costs a significant amount of energy, resulting in a high reaction barrier on Pd (111). To quantitatively rationalize the difference between the two reaction barriers, an equation was proposed to decompose the reaction barrier of CO + O → CO_2:

$$E_a = E_O + E_{CO} + E_{surf} + E_m$$

where E_a is the reaction barrier, E_O and E_{CO} are the energies required to activate O and CO from the initial state to the TS, respectively. E_{surf} is the energy associated with surface relaxation during the reaction, and E_m is a mixing term that includes the O—CO bond formation energy and an energy due to bonding competition [44]. E_{surf} is found to be small (~0.1 eV) at the interface and on Pd (111) surface. E_m is metal dependent and is related to the TS structure. As shown in the Figs. 7c and 9 c, the transition state structures with the lowest reaction barriers on Pt/Co_3O_4 and Pd (111) are similar and E_m is expected to be similar on both surfaces. Therefore, Ea mainly originates from E_O and E_{CO}. E_O represents the energy of O movement from the hollow site toward the bridge site. The value of E_O on Pt/Co_3O_4 is similar with that on Pd (111) surface. Consequently, the major difference in E_a between the interface and Pd (111) surface lies in the second term E_{CO}. The calculated E_{CO} is less than 0.1 eV for the interface but quite large (0.99 eV) for Pd (111) surface, resulting in the different energy barriers between the interface and Pd (111) surface.

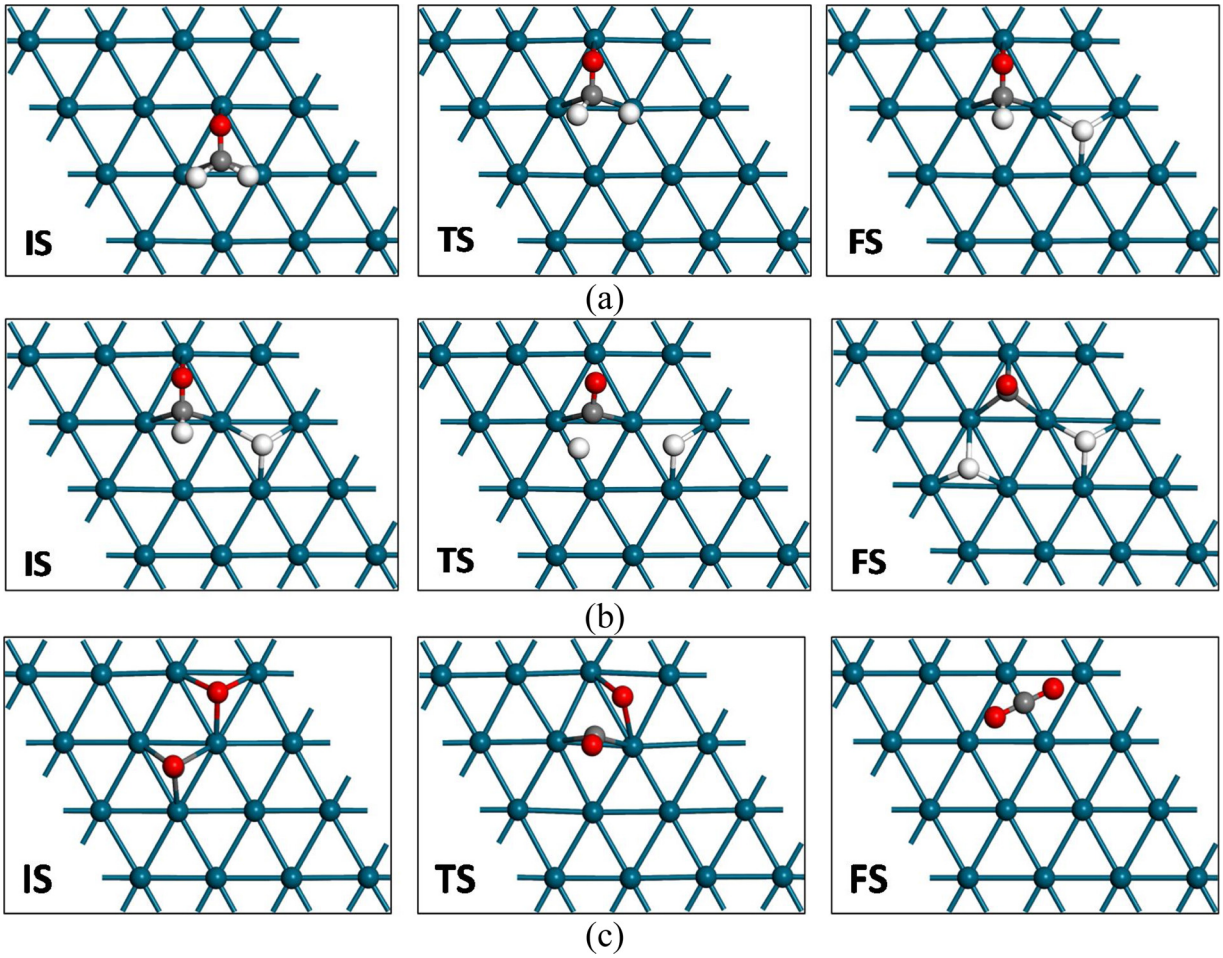


Fig. 9. The proposed reaction path on Pd (111) surface. (a) HCHO → CHO + H, (b) CHO → CO + H, (c) CO + O → CO_2. (IS, TS, FS stand for initial state, transition state, final state).

Table 1
The prefactor, activation energy, forward and backward rate constants of each elementary reaction step of HCHO oxidation at the Pd/Co_3O_4 interface.

Reaction steps	prefactor	activation energy (kJ/mol)	forward rate constants	backward rate constants
HCHO + * → HCHO*		0	2.52e9	2.55e4
O_2 + * →O_2*		0	2.66e9	1.14e2
O_2* → O* + O*	1.00e13	69.2	9.73e12	9.20e12
HCHO* → CHO* + H*	1.00e13	52.9	9.79e12	9.69e12
CHO* → CO* + H*	1.00e13	56.7	9.78e12	9.86e12
H* + O* → OH*	1.00e13	66.3	9.74e12	9.65e12
H* + OH* → H_2O + *	1.00e13	86.5	9.66e12	9.46e12
CO* + O * → CO_2 + *	1.00e13	48.1	9.81e12	9.09e12

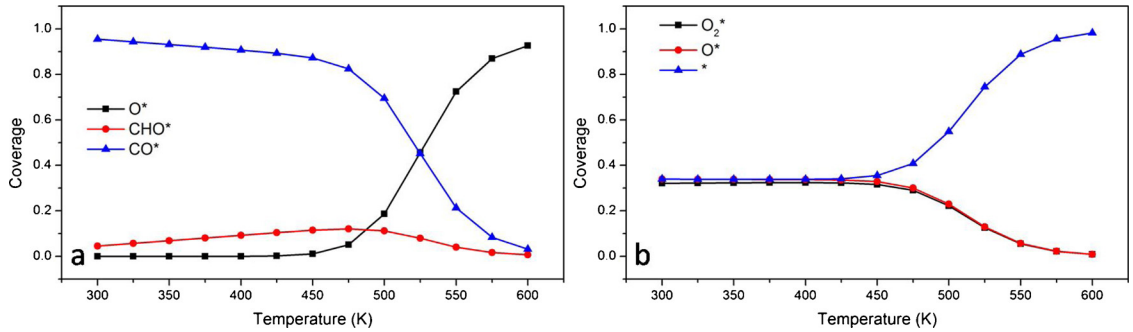


Fig. 10. The surface coverage of intermediates as a function of temperature on HCHO oxidation (a) on Pd (111) surface, (b) at Pd/Co_3O_4 interface. The intermediates with a surface coverage of 0 is not given.

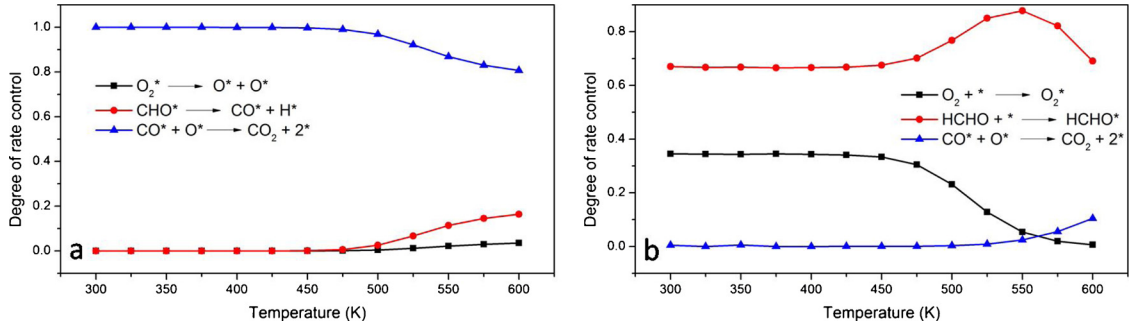


Fig. 11. The degree of rate control for element steps as a function of temperature on HCHO oxidation (a) on Pd (111) surface, (b) at Pd/Co_3O_4 interface.

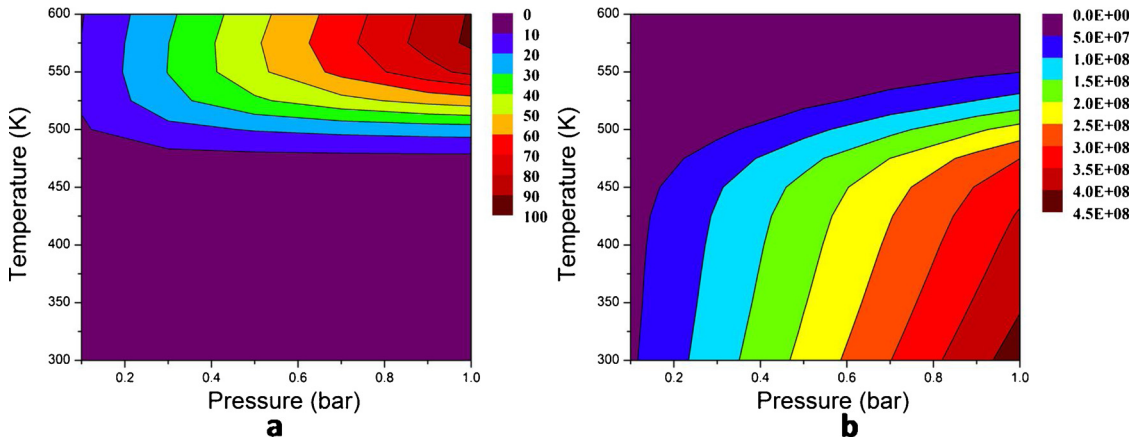


Fig. 12. The overall reaction rate (s^{-1}) as a function of temperature (K) and gas total pressure (bar) (a) on Pd (111) surface, (b) at Pd/Co_3O_4 interface.

3.4. Microkinetics study

HCHO oxidation reaction cycle at Pd/Co_3O_4 interface can be described as Langmuir–Hinshelwood mechanism: HCHO and O_2 are adsorbed on the surface, and the adsorbed O_2 dissociates into atomic O* species. Then, HCHO is decomposed to CHO and CO, and CO molecule combines with O* to form CO_2. Finally, the dissociated H atoms combines with another O* to form H_2O. The desorption process is omitted because the formed H_2O and CO_2 go into gas phase directly. Because the energy barrier of dissociated H migration on Pd surface is so small, it is reasonable to assume that the movement of H on Pd surface has less effect on the whole reaction cycle. The prefactors, activation energies, forward and backward rate constants of each elementary reaction step of HCHO oxidation at the Pd/Co_3O_4 interface is shown in Table 1. For surface reactions, there is no significant change in the entropy and consequently the pre-exponential factor of the rate constants is typically in the order of 10^{13}. Furthermore, chemisorption is a non-activated process and the activation energy equals zero. The production rate of H_2O and CO_2 at room temperature (300 K) were compared. The results show that the production rate at Pd/Co_3O_4 interface ($4.27 \times 10^8\ s^{-1}$) is many orders of magnitude higher than that on Pd (111) surface ($7.89 \times 10^{-7}\ s^{-1}$), indicating that Pd/Co_3O_4 interface largely improves the overall HCHO catalytic activity, especially at room temperature. Fig. 10 shows surface coverage of the intermediates as a function of temperature. On Pd (111) surface, the adsorbed CO molecules occupy the main active sites. With the increase of temperature, the coverage of CO declines gradually. The reason is that the high energy barrier of CO oxidation limits the conversion of CO to CO_2. When the temperature rises, the rate of CO + O → CO_2 increases and CO molecules are consumed. Thus, the coverage of CO decreases. It can be also found that the coverage of dissociation O increases with the temperature increasing, which may contribute to the rapid adsorption and dissociation of O_2 at high temperature. In contrast, there is no CO molecule at Pd/Co_3O_4 interface (Fig. 10b). The interface interaction reduces the energy barrier of CO oxidation and CO molecule can be completely converted to CO_2 at low temperature. The interface is partly covered with adsorbed O_2 molecule and dissociation O species. The initial concentration of HCHO and O_2 is set as 1:1. At low temperature, the adsorption of HCHO is weaker than that of O_2. Thus, there are O_2 molecules left at the interface and HCHO molecules are completely consumed. With the increasing of temperature, the desorption of HCHO and O_2 increases rapidly, and the adsorbed HCHO and O_2 can be completely converted to H_2O and CO_2. There is only the active sites left at the interface after the desorption of H_2O and CO_2.

The degree of rate control (X_{RC}) is calculated for each step and the results are shown Fig. 11. On Pd (111) surface, the step of CO + O → CO_2 has the largest X_{RC} value in the whole temperature range, indicating that CO oxidation remarkably influences HCHO oxidation reaction. With the increasing of temperature, the oxidation rate of CO improves and the influence on reaction cycle is weak. By contrast, CO oxidation has less effect on the whole HCHO oxidation due to its low energy barrier at the interface. Therefore, the big X_{RC} values is generated form the step of HCHO and O_2 adsorption at Pd/Co_3O_4 interface. Although the formation energy barrier of water is 0.90 eV, the X_{RC} value of step H* + OH* → H_2O + * is only 0.05. It shows that the formation of water has little effect on the whole reaction. Fig. 12 shows the reaction rate with respect to the increase in the temperature and pressure (the total pressure of gas phase contained a mixture of HCHO and O_2 in 1:1 ratio). When the temperature is below 500 K, the overall reaction rate of HCHO oxidation on Pd (111) surface is lower than 10 s^{-1}. When the temperature is above 500 K, the whole reaction rate rises with the increasing of pressure and temperature. For HCHO oxidation at Pd/Co_3O_4 interface (Fig. 12b), the whole reaction rate increases when the pressure rises. However, the reaction rate declines with the increase of temperature. The reason is that O_2 desorption rate increases when the temperature rises.

4. Conclusion

This work is aimed at obtaining a comprehensive understanding of interfacial effects on HCHO catalytic reactions over Pd/Co_3O_4. Compared with pure Co_3O_4 (110) surface, Pd plays a role of active site in HCHO oxidation. The energy barrier of C–H cleavage on Pd sites is lower than that on Co_3O_4 (110) surface. CO molecule can react with adsorbed O_2 to form CO_2 instead of lattice oxygen atoms. Compared with Pd (111) surface, Pd/Co_3O_4 interface has less effect on C–H bond cleavage and O_2 dissociation. The main influence caused by the interface on HCHO oxidation is the step of $CO + O \rightarrow CO_2$. Firstly, the interaction between noble Pd and Co_3O_4 results in the migration of electron from Pd to Co_3O_4, which contributes to HCHO and O_2 adsorption. Secondly, the interface effect decreases the energy barrier of CO oxidation. The results of barrier decomposition analysis show that the high energy barrier of CO oxidation on Pd (111) surface originates from the separation of CO molecule from the surface. Thus, CO at the interface is easier to be activated and forms CO_2.

Microkinetics study of HCHO catalytic oxidation indicates that CO oxidation on Pd (111) surface is the rate limiting step, and the catalyst surface is covered with CO molecules at low temperature. At Pd/Co_3O_4 interface, CO oxidation is easy to achieve due to the low energy barrier. The adsorption of HCHO and O_2 is the constraint. HCHO can completely convert into CO_2 and H_2O with the temperature increasing. The production rate of CO_2 and H_2O at Pd/Co_3O_4 interface is fifteen orders of magnitude higher than that on Pd (111) surface at room temperature, indicating that Pd/Co_3O_4 interface can enhance HCHO catalytic activity.

Conflict of interest

The authors declare no competing financial interest.

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